Comprehensive 18-Week Syllabus Outline for Advanced Computer Science Paradigm

An intensive blueprint constructed optimized for university platforms targeting theory + practical.

Week 1: Foundations of Mobile Testing & Core Paradigms

Topics:
- Definition of mobile testing across phones and tablets
- Visual, behavioral, and crash testing parameters
- Key differences between mobile and desktop environments
- Real-world case study: Banking application failure analysis

Objectives:
- Understand the core definition and necessity of mobile testing
- Analyze why mobile apps fail on older devices compared to development environments

Tasks:
- Analyze a simulated banking application on multiple virtual screen sizes
- Document visual and behavioral discrepancies

Week 2: Android Fragmentation and Device Diversity

Topics:
- The concept of Android fragmentation
- Managing 24,000+ Android device models
- Active Android version distribution (8+ active versions)
- Screen size variations (4 to 7 inches) and layout adaptation
- Manufacturer-specific OS modifications (Samsung, Xiaomi, OnePlus, Huawei)

Objectives:
- Identify the challenges posed by Android fragmentation
- Formulate a device selection matrix for testing target markets

Tasks:
- Create a device matrix targeting top manufacturers and screen sizes
- Analyze layout responsiveness across different aspect ratios

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Week 3: iOS Testing Ecosystem and Platform Comparisons

Topics:
- iOS testing challenges despite fewer device types
- Coexistence of multiple active iOS versions (15, 16, 17)
- Form factor differences (iPhone SE vs. iPhone 15 Pro Max)
- Mac OS hardware requirements for iOS testing
- Strict Apple App Store submission rules and pre-release quality gates
- Comparative analysis: Android vs. iOS testing strategies

Objectives:

- Contrast the testing challenges of iOS with Android
- Understand the hardware and software constraints of the iOS ecosystem

Tasks:
- Draft a comparative testing strategy document for a cross-platform app
- Simulate an App Store compliance review checklist

Week 4: Environmental and Network Testing Challenges

Topics:
- Network speed variability (5G, 3G, offline states)
- Battery consumption and performance profiling
- Memory leaks and resource-heavy application behavior
- User retention impacts of slow or battery-draining apps

Objectives:
- Design test cases that simulate poor network conditions
- Measure and analyze application battery and memory usage

Tasks:
- Use network throttling tools to test app behavior under 3G and offline conditions
- Profile an application's battery usage during intensive tasks

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Week 5: Interactive and System-Level Interruptions

Topics:
- Handling system interruptions (phone calls, text messages, pop-ups)
- App recovery states and state preservation
- Permissions and privacy testing (camera, location, contacts)
- Handling user permission denials
- Touch and gestures automation challenges (swipe, pinch, zoom, long-press)
- Screen orientation transitions (portrait vs. landscape layout checks)

Objectives:
- Implement test scenarios for unexpected system interruptions
- Verify application layout and state preservation during screen rotation

Tasks:
- Write test cases for permission denial recovery
- Manually verify gesture-heavy interfaces and document failure points during orientation changes

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Week 6: Introduction to Test Automation and Appium

Topics:
- The role of automation in mobile testing
- What is Appium: Free, open-source automation framework
- Cross-platform capabilities (one test script for Android and iOS)
- No-code-modification philosophy of Appium
- Language independence (Java, Python, JavaScript, Ruby)
- The 'Robot Hand' analogy for automated interactions

Objectives:
- Explain the core benefits of using Appium for mobile automation

- Understand the philosophy of non-intrusive testing

Tasks:
- Research and present a comparison of Appium with other mobile automation tools
- Map out the conceptual flow of an automated tap action

Week 7: Appium Architecture and the WebDriver Protocol

Topics:
- The three-part Appium architecture
- The Client/Test Script role (The Customer)
- The Appium Server role (The Waiter)
- The Mobile Device/Driver role (The Kitchen)
- The WebDriver Protocol standard (Selenium alignment)
- How commands are translated and executed on devices

Objectives:
- Deconstruct the Appium communication lifecycle
- Explain how the WebDriver protocol enables cross-platform automation

Tasks:
- Draw a sequence diagram of an Appium command execution from script to device
- Start the Appium server via terminal and inspect the logs

Week 8: Appium Environment Setup and Driver Installation

Topics:
- Node.js installation and verification
- Global Appium installation via npm
- Platform-specific driver installation (uiautomator2 for Android, xcuitest for iOS)
- The role of drivers in platform communication
- Troubleshooting common installation errors

Objectives:
- Successfully configure a local Appium server environment
- Install and verify necessary platform drivers

Tasks:
- Install Node.js and Appium globally on a local machine
- Run terminal commands to install and list active Appium drivers

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Week 9: Platform SDKs and Path Configurations

Topics:
- Android Studio installation and SDK manager
- Setting the ANDROID_HOME environment variable
- Xcode installation requirements for iOS
- Configuring system paths for terminal execution
- Verifying environment readiness

Objectives:
- Configure system environment variables for Android and iOS development tools
- Verify SDK path accessibility from the command line

Tasks:

- Set up ANDROID_HOME and update system PATH variables
- Run diagnostic checks to ensure Android SDK tools are recognized

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Week 10: Writing and Executing Your First Appium Test

Topics:
- Importing the Appium webdriver in Python
- Defining Desired Capabilities / Appium Options
- Specifying application paths (.apk) and device names
- Establishing a remote connection to the Appium server (port 4723)
- Locating elements using IDs
- Executing click actions and closing the driver session

Objectives:
- Write a complete, executable Appium automation script in Python
- Establish a successful connection to a running Appium server

Tasks:
- Write a Python script to launch a sample mobile application
- Locate a login button by ID, perform a click, and terminate the session safely

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Week 11: Emulators vs. Real Devices in Automation

Topics:
- Pros and cons of Real Devices (accuracy, network, cost, management)
- Pros and cons of Emulators/Simulators (cost, setup, scalability, hardware limitations)
- Hybrid testing strategies: Emulators for development, Real Devices for release
- Best practices for daily test execution pipelines

Objectives:
- Evaluate when to use emulators versus real devices in a testing lifecycle
- Implement a hybrid testing strategy for a development sprint

Tasks:
- Configure and launch an Android Virtual Device (AVD)
- Run the Week 10 script on both an emulator and a connected real device, comparing execution times

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Week 12: Object-Oriented Programming (OOP) Concepts Review

Topics:
- Review of OOP definitions and core pillars
- Encapsulation: Hiding internal details (TV remote analogy)
- Inheritance: Parent-child class relationships (Animal-Dog analogy)
- Polymorphism: Dynamic method execution (Shape-Circle-Square analogy)
- Abstraction: Simplifying interfaces (Driving a car analogy)

Objectives:
- Articulate the four main pillars of OOP
- Relate OOP design patterns to software testability

Tasks:
- Write a simple class hierarchy in Python or Java demonstrating inheritance and polymorphism

• Identify encapsulation boundaries in a provided code sample

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Week 13: Testing Object-Oriented Systems: Inheritance Pitfalls

Topics:
• Introduction to testing OO systems
• The 'Changing Parent Breaks All Children' pitfall
• How modifications in parent classes affect untouched child classes
• Regression testing strategies for class hierarchies
• Isolating parent and child behaviors

Objectives:
• Identify potential failure points in inherited codebases
• Design regression test suites specifically targeting parent class modifications

Tasks:
• Modify a parent class method and trace the cascading failures in its child classes
• Write unit tests to catch unintended side effects in child classes

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Week 14: Deep Inheritance Chains and Overriding Surprises

Topics:
• The complexity of deep inheritance chains (A -> B -> C -> D)
• Tracing bugs through multi-level hierarchies (Vehicle -> Car -> ElectricCar -> Tesla)
• Method overriding pitfalls (Dog.speak overriding Animal.speak returning null)
• Strategies for isolating classes using mock objects

Objectives:
• Trace and debug errors across deep inheritance hierarchies
• Apply mocking techniques to isolate individual classes during testing

Tasks:
• Create a mock object to isolate a subclass from its deep parent chain
• Debug a simulated runtime crash caused by an overridden method returning unexpected types

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Week 15: Testing Polymorphism: Pitfalls and Late Binding

Topics:
• The definition of polymorphism in testing contexts
• Pitfall 1: The 'Testing Only One Type' fallacy
• Pitfall 2: Late binding bugs (runtime method selection)
• Pitfall 3: Missing methods in child classes causing runtime crashes
• Why compile-time checks fail to catch polymorphism bugs

Objectives:
• Analyze the unique risks associated with polymorphic methods
• Explain how late binding delays bug detection to runtime

Tasks:
• Write a test suite that exposes a late-binding runtime crash in a polymorphic system
• Document a case where testing a single type failed to catch a subclass bug

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Week 16: Advanced Polymorphism Testing Strategies

Topics:
- The Golden Rule: Test every class that uses a polymorphic method
- Designing independent unit tests for each subclass
- Using mock objects to replace real dependencies
- Ensuring complete coverage of all polymorphic variants

Objectives:
- Apply the Golden Rule of polymorphism testing to a multi-class system
- Implement isolated unit tests using mock frameworks

Tasks:
- Refactor a shared test suite into separate, isolated tests for three distinct subclasses
- Implement mock dependencies to test a polymorphic class in isolation

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Week 17: Runtime Behavior and Integration Testing

Topics:
- Writing tests that execute apps with diverse object types
- Verifying dynamic runtime outputs
- Transitioning from unit tests to integration testing
- How polymorphism bugs manifest during multi-class interactions

Objectives:
- Design integration tests that verify class interactions
- Validate dynamic runtime behavior under varied inputs

Tasks:
- Build an integration test suite that runs a workflow using multiple polymorphic objects
- Analyze interaction logs to verify correct dynamic binding

Week 18: Capstone Synthesis, Golden Rules, and Course Wrap-Up

Topics:
- Synthesizing Mobile Testing and OO Testing paradigms
- Review of the Golden Rules of testing
- Designing a comprehensive test plan for a mobile OO application
- Course review and future trends in mobile/OO automation

Objectives:
- Synthesize mobile automation and object-oriented testing principles into a unified strategy
- Present a comprehensive test plan for a complex application

Tasks:
- Develop and present a complete test plan and automated test suite for a mobile app utilizing polymorphic structures
- Complete the final course assessment
